

CAIRNS, Australia

WMO: 942870

Lat: 16.8736S

Lon: 145.7450E

Elev: 27

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	55.9	58.9	43.4	41.5	69.7	46.9	47.5	67.8	24.4	75.5	22.8	75.4	4.6	280	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	91.2	77.0	89.6	76.7	88.2	76.5	80.5	86.7	79.7	85.8	79.0	85.0	12.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	150.3	84.2	78.0	145.8	83.4	77.2	141.8	82.6	44.3	87.4	43.4	86.2	42.7	85.3	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	20.5	18.7	51.5	96.5	2.7	3.1	49.6	98.7	48.0	100.6	46.5	102.3	44.5	104.6
			49.1	82.7	1.9	1.0	47.7	83.5	46.6	84.0	45.5	84.6	44.1	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	81.6	81.5	80.3	78.1	75.0	72.2	70.6	71.4	74.2	77.3	79.6	81.4	(6)
	DBStd	4.72	2.20	2.32	2.06	2.31	2.64	3.22	3.31	2.96	2.73	2.41	2.07	2.37	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	6	0	0	0	0	0	2	3	1	0	0	0	0	(9)
	CDD50	9824	980	883	940	844	777	665	640	664	727	845	887	972	(10)
	CDD65	4356	515	463	475	394	312	217	178	201	277	380	437	507	(11)
	CDH74	34457	5307	4665	4299	2808	1543	734	570	858	1671	2818	3934	5250	(12)
	CDH80	9718	1839	1567	1258	662	208	43	22	66	279	711	1199	1864	(13)
(14) Wind	WSAvg	9.8	8.3	8.2	9.8	11.2	11.0	11.0	10.9	10.4	9.6	9.6	9.1	8.6	(14)
(15) Precipitation	PrecAvg	79.2	15.4	17.9	16.0	8.0	3.5	1.8	1.2	0.9	1.2	1.8	3.7	7.1	(15)
	PrecMax	124.8	55.8	51.5	43.8	25.0	12.7	7.1	5.7	3.2	4.0	8.0	14.6	36.2	(16)
	PrecMin	28.3	3.4	1.2	1.1	0.5	0.1	0.1	0.1	0.0	0.0	0.1	0.1	1.1	(17)
	PrecStd	21.3	10.5	10.7	10.5	5.2	2.6	1.5	1.0	0.8	1.3	1.8	3.4	6.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	94.0	91.0	88.1	85.5	82.8	82.2	83.8	86.1	88.2	90.0	92.8	(19)
		MCWB	77.6	78.4	77.6	74.6	73.2	71.4	69.9	68.9	69.2	72.8	74.8	76.9	(20)
	2%	DB	90.3	90.6	88.9	86.4	83.8	81.0	80.5	81.9	84.3	86.4	88.1	90.1	(21)
		MCWB	77.3	77.8	76.8	74.0	72.2	70.1	68.6	68.3	69.6	72.7	74.2	76.2	(22)
	5%	DB	88.9	89.1	87.7	85.5	82.4	79.8	78.9	80.4	82.8	85.2	87.0	89.1	(23)
		MCWB	77.2	77.8	76.6	73.8	71.3	69.3	67.2	67.5	69.1	72.2	73.9	76.2	(24)
	10%	DB	87.6	87.6	86.2	84.1	80.8	78.5	77.2	78.8	81.7	84.0	85.8	87.7	(25)
		MCWB	77.3	77.8	76.1	73.4	70.6	68.5	66.3	66.4	68.9	71.8	73.8	76.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.4	82.0	80.9	78.9	76.3	74.6	72.9	73.3	75.6	77.5	79.5	81.0	(27)
		MCDB	87.2	88.1	86.9	83.8	82.2	79.6	77.3	78.2	81.6	83.8	85.5	87.4	(28)
	2%	WB	80.2	80.3	79.6	77.6	74.7	72.7	71.1	71.3	73.8	76.3	78.0	79.8	(29)
		MCDB	86.3	86.5	85.4	82.8	80.0	77.8	76.9	77.7	79.9	82.5	84.2	86.3	(30)
	5%	WB	79.4	79.4	78.7	76.4	73.6	71.6	70.0	69.8	72.4	75.1	77.0	78.9	(31)
		MCDB	85.4	85.5	84.2	81.8	79.0	76.8	75.5	76.5	79.0	81.6	83.2	85.4	(32)
	10%	WB	78.7	78.7	77.8	75.4	72.5	70.3	68.9	68.7	71.1	73.9	75.9	78.0	(33)
		MCDB	84.6	84.7	83.1	80.7	78.1	75.7	74.6	75.9	78.2	80.7	82.5	84.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	11.8	11.6	12.0	12.5	12.8	14.4	15.5	15.8	14.2	13.4	12.8	(35)
		MCDBR	14.3	14.3	13.8	14.1	13.6	13.8	14.9	17.0	16.8	14.9	14.6	14.7	(36)
	5% WB	MCWBR	5.8	5.9	5.5	5.8	5.9	6.4	6.5	7.0	6.7	5.7	5.5	5.4	(37)
		MCWBR	12.3	12.1	11.4	10.8	11.1	11.2	12.0	13.9	14.0	12.9	12.3	12.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.393	0.394	0.392	0.379	0.366	0.359	0.342	0.349	0.377	0.395	0.395	0.409	(40)
		taud	2.574	2.573	2.566	2.582	2.588	2.591	2.624	2.589	2.488	2.463	2.502	2.490	(41)
	Ebn at Noon		302	298	292	285	278	274	283	290	292	295	299	297	(42)
		Edh at Noon	34	34	33	30	28	27	27	30	35	37	36	37	(43)
(44) All-Sky Solar Radiation	RadAvg		1876	1801	1664	1531	1350	1206	1324	1607	1937	2103	2150	2063	(44)
	RadStd		156	207	145	150	118	112	99	124	102	137	171	230	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.56	N/A	+0.36	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page